AI Resume Analyzer and Job Recommendation System

Major Project 1 | Project Summary

Project Overview

This project is a Python-based application that analyzes a resume, identifies listed technical and soft skills, and recommends suitable job roles. It is designed as a decision-support tool for exploring job-role alignment and possible learning areas—not as a hiring decision system.

Objectives

- Extract text from PDF and DOCX resumes.
- Identify skills using a curated skill dictionary.
- Compare identified skills with requirements for five job roles.
- Display the top three role recommendations, estimated match scores, missing skills, and a basic learning roadmap.
- Provide a downloadable text report while minimizing unnecessary resume storage.

Technologies

Python • Streamlit • pandas • scikit-learn • pypdf • python-docx • Matplotlib (notebook visualization)

Methodology / Workflow

1. Upload	2. Extract & clean	3. Identify skills
PDF/DOCX input	Read resume text	Match dictionary terms
4. Compare	**5. Recommend**	**6. Report**
TF-IDF / cosine similarity	Rank top three roles	Show chart & learning gaps

Main Features

• PDF/DOCX upload with a 10 MB size limit • Skill identification • Top-three job recommendations • Match-score chart • Target-role selector • Missing-skill list and learning roadmap • Downloadable analysis report

Evaluation and Limitations

The application should be evaluated using test cases for PDF/DOCX extraction, unsupported files, file size, skill detection, ranking, chart display, missing skills, report download, and unreadable documents. Record actual outcomes in the test sheet. Match scores are similarity estimates, not probabilities of employment or measures of a person's ability. Results depend on resume text and dictionary coverage; false matches and missed skills are possible. The tool should not use protected attributes to rank candidates.

Expected Deliverables

Working Streamlit app • Source code and datasets • Core Google Colab notebook • Test sheet with observed results • README and workflow diagram • This one-page project summary • Student-recorded demonstration video

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